

1. Administrative & Study Identification

Full Study Title: Substudy 06D: Combination Therapies in Second Line (2L) Gastroesophageal Adenocarcinoma (MK-3475-06D/KEYMAKER-U06) **Short Title/Acronym:** KEYMAKER-U06 Substudy 06D **Protocol Number:** MK-3475-06D / 3475-06D **NCT Number:** NCT06445972 **Posting ID:** 737280407558717956 **Trial Status:** Recruiting (Currently recruiting participants) **Sponsor/Company Name:** Merck (Merck Sharp & Dohme LLC) **Investigational Product:** Sacituzumab tirumotecan (MK-2870) plus paclitaxel; HER3-DXd plus ramucirumab; Ramucirumab plus paclitaxel (active comparator) **Phase of Development:** Phase 1/2 **Medical Monitor:** Merck Sharp & Dohme LLC Clinical Director (Contact Information on file)

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the safety and tolerability of the study interventions during the safety lead-in phase, specifically assessing Dose Limiting Toxicities (DLTs), adverse events (AEs), and discontinuations due to AEs, as well as Objective Response Rate (ORR) for the efficacy phase. **Secondary Objectives:** To evaluate Progression-Free Survival (PFS), Duration of Response (DOR), and Overall Survival (OS).

2.2 Background & Rationale Treatment options for patients with advanced or metastatic gastroesophageal adenocarcinoma (GEA), including esophageal neoplasms and cancer of the gastroesophageal junction, are limited, and the overall prognosis remains poor. This study addresses this unmet medical need by evaluating the safety and efficacy of novel antibody-drug conjugates (ADCs) such as the TROP2 ADC sacituzumab tirumotecan (MK-2870) and HER3-DXd in combination with standard-of-care chemotherapy (paclitaxel) or targeted agents (ramucirumab) as second-line therapy for patients who have failed one prior line of therapy.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional (Umbrella Platform Study) **Control Type:** Active Comparator (Ramucirumab plus paclitaxel) **Blinding:** Open-label **Randomization:** 1:1 Randomized (in the efficacy phase)

3.2 Planned Enrollment & Duration Target \$N\$: 90 participants (Targeted Enrollment) **Number of Sites:** Multicenter (Global, including sites in the US, Brazil, Chile, Taiwan, and China)

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 18 years with an Eastern Cooperative Oncology Group (ECOG) performance status of 0 or 1, and a life expectancy of at least 3 months. 2. Histologically and/or cytologically confirmed diagnosis of previously treated, second-line (2L) advanced/metastatic gastric adenocarcinoma, gastroesophageal junction (GEJ) adenocarcinoma, or esophageal adenocarcinoma. 3. Documented objective radiographic or clinical disease progression during or after 1L therapy. 4. Must provide an archival tumor tissue sample or most recently obtained core, incisional, or excisional biopsy of a tumor lesion. Participants with HER2/neu positive tumors are excluded, but participants with unknown HER2 status are eligible. 5. Hepatitis B (HBsAg positive), Hepatitis C (HCV), and HIV-infected participants are eligible provided their viral loads are undetectable or well-controlled on respective antiviral/antiretroviral therapies per protocol.

4.2 Exclusion Criteria 1. Squamous cell or undifferentiated gastroesophageal cancer, or known active central nervous system (CNS) metastases. 2. Has experienced weight loss $>20\%$ over 3 months before the first dose of study intervention. 3. History of documented severe dry eye syndrome, severe Meibomian gland disease, blepharitis, or severe corneal disease. 4. Has Grade ≥ 2 peripheral neuropathy. 5. Active inflammatory bowel disease (IBD) requiring immunosuppressive medication or previous history of IBD, serious or nonhealing wound, peptic ulcer, or bone fracture within 28 days prior to randomization. 6. History of (noninfectious) pneumonitis/interstitial lung disease (ILD) that required steroids, or current pneumonitis/ILD. 7. Severe hypersensitivity (Grade ≥ 3) to MK-2870, HER3-DXd, or any of their excipients.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: * *Investigational Arm 1:* Sacituzumab tirumotecan (3 or 4 mg/kg IV in safety lead-in; or on days 1, 15, and 29 of each 6-week cycle) plus paclitaxel (ATC Code: L01CD01; Trade names: Abraxane, Onxol, Taxol, Paccal). * *Investigational Arm 2:* HER3-DXd plus ramucirumab. * *Comparator Arm:* Ramucirumab (8 mg/kg IV on days 1 and 15) plus paclitaxel (80 mg/m² IV on days 1, 8, and 15) of each 4-week cycle. **Route of Administration:** Intravenous (IV) **Duration of Treatment:** Until documented disease progression, unacceptable toxicity, or study withdrawal.

5.2 Concomitant & Prohibited Therapy Permitted: Routine clinical care, prophylactic medications, and endocrine-related AEs adequately treated with hormone replacement. **Prohibited:** Received a live or live-attenuated vaccine within 30 days prior to the first dose; prior radiotherapy within 2 weeks requiring corticosteroids; use of another investigational agent within 4 weeks.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Tumor Biopsy (HER2 status assessment), Medical History, Baseline Imaging
Baseline	Day 0	Randomization, First Dose Administration
Interim	Every 4 to 6 weeks	Safety Labs, Efficacy Assessment (Imaging per RECIST v1.1), AE Evaluation
End of Study	Follow-up	Final Treatment Visit, Survival Follow-up, ADA Immunogenicity Testing

7. Outcome Measures (Endpoints)

7.1 Primary Endpoints * **Safety Lead-in Phase:** 1. Percentage of Participants who Experience Dose Limiting Toxicities (DLTs). 2. Percentage of Participants who Experience an Adverse Event (AE). 3. Percentage of Participants who Discontinue Study Intervention Due to an AE. * **Efficacy Phase:** Objective Response Rate (ORR) per RECIST v1.1. **Measurement Method:** Continuous AE/DLT recording, and Blinded Independent Central Review (BICR) of imaging assessments.

7.2 Secondary & Exploratory Endpoints * Progression-Free Survival (PFS) per RECIST v1.1 by BICR. * Duration of Response (DOR) per RECIST v1.1 by BICR. * Overall Survival (OS).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous AE monitoring graded by standard oncology toxicity criteria (NCI CTCAE v5.0); specific evaluation of immunogenicity (Anti-Drug Antibody [ADA] responses) to MK-2870 and HER3-DXd using validated assays. **Serious Adverse Events (SAEs):** Standard 24-hour reporting protocols for serious or life-threatening adverse events. **Data Monitoring Committee (DMC):** Internal/External safety review committee per MSD umbrella protocol operating guidelines.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for efficacy endpoints; Safety Set for AE/SAE evaluations. **Sample Size**

Justification: This is an estimation study; no formal hypothesis testing will be performed for the 90 targeted participants.
Handling of Missing Data: Standard oncology data handling, incorporating censoring for time-to-event endpoints (PFS, OS).

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.
Monitoring Plan: Periodic onsite and remote monitoring visits to ensure protocol adherence and safety compliance. **Audit Trail:** Secure eCRF locking, source data verification, and data clarification forms.

11. Study Identifiers & Governance

Primary ID: {{MK-3475-06D}} **Registry Number:** {{NCT06445972}}
Posting ID: {{737280407558717956}} **Sponsor/Lead Organization:** {{Merck Sharp & Dohme LLC}} **Study Title:** {{KEYMAKER-U06 Substudy 06D}} **Official Regulatory Title:** {{A Phase 1/2 Open-Label, Umbrella Platform Design Study to Evaluate the Safety and Efficacy of MK-2870 Plus Paclitaxel as the Second-Line Treatment of Participants With Advanced/Metastatic Gastroesophageal Adenocarcinoma: Substudy 06D}}

12. Clinical Framework (GEA Specific)

Disease Areas: {{Gastroesophageal Junction, Gastroesophageal Adenocarcinoma, Esophageal Neoplasms, Esophageal Cancer}}
Diagnosis Threshold: {{Documented progression during or after 1L therapy; known HER2-positive tumors excluded}} **Medical Methodology:** {{TROP2 or HER3 targeted Antibody-Drug Conjugates (ADCs) combined with systemic therapy}} **Optimization Target:** {{Objective Response Rate (ORR) and extending Progression-Free Survival (PFS)}}

13. Study Procedures & Methodology

Management Pathway: {{Oncology clinical pathway for 2L GEA management}} **Education Protocol (HCP):** {{Protocol-specific dosing, infusion procedures, and AESI (e.g., pneumonitis/ILD, ocular toxicities) monitoring guidelines}} **Education Protocol (Patient):** {{Informed consent, toxicity symptom recognition, and AE reporting}} **Quality Audit Mechanism:** {{Blinded Independent Central Review (BICR) of radiographic imaging}}

14. Observation & Follow-Up Schedule

Total Duration: {{Follow-up extending until disease progression, death, or trial completion}} **Onsite Visit Cadence:** {{Every 4 to 6 weeks matching respective IV treatment cycles}} **Remote Monitoring Frequency:** {{Standard long-term survival follow-up intervals}} **Checklist Review Interval:** {{Prior to each treatment cycle dosing}}

15. Sign-off & Approval

Role	Name	Signature	Date		:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]					
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]					Lead
Statistician	[Name]	_____	[DD-MMM-YYYY]					